Son of Surya

GUIDE
R3
FUNDAMENTALS

FREE FOUNDATION GUIDE | BEGINNER

Sun, Site & Shade

Irradiance, peak sun hours, tilt, direction and shadows — how to judge an Indian roof or plot before a single panel is installed.

W/m²	PSH	°	S	9–3
IRRADIANCE	SUN HOURS	TILT	DIRECTION	SHADE RULE

“Understand your solar before you buy your solar.”

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R3 About This Guide

Guide	**R3 — Sun, Site & Shade**: irradiance, peak sun hours, tilt and orientation in India
Category · Level · Price	Solar Fundamentals · Beginner → Foundation · **Free**
For	Homeowners, farmers and building owners checking whether their roof or land suits solar
Prerequisites	R1 Electricity Basics
Read next	SOS-01 Solar Starter · SOS-06 Rooftop Sizing · R7 Structure & Mounting · R8 Coastal & Harsh Climates

The problem

Two identical systems can produce very different units. The difference is sunlight: where you are, which way the panels face, how steeply they tilt, and what casts shadows on them.

The knowledge gap

People judge a roof at noon in summer, when shadows are shortest. The real test is **9 am to 3 pm in December**, when the sun is lowest and shadows are longest.

The decision

After reading, you can judge whether your site is suitable, estimate how many units it can produce, and spot shade problems before anyone installs anything.

The action

Do the 20-minute site walk in Section 13, measure your shade-free area, and use your region's planning figure to estimate yearly units.

01 What You Will Learn

- [ ] Irradiance (W/m²) vs irradiation (kWh/m²/day) — and why "peak sun hours" is the number that matters
- [ ] How the sun moves across Indian skies through the year
- [ ] Which direction and tilt suit Indian rooftops, and what you lose when you compromise
- [ ] How to predict shadows from tanks, walls, trees and neighbouring buildings
- [ ] How much shade-free area you need per kWp
- [ ] Regional planning figures for estimating yearly units

02 Why This Matters

A panel's rating (Wp) is fixed. Its **units** are not — they depend on how much sunlight actually reaches it. A shadow from a water tank on a few panels can pull down a whole string's output. A badly chosen direction can cost

output every day for 25 years. These decisions are cheap to get right before installation and expensive to fix after.

WHAT CHANGES IN INDIA?

India lies roughly between 8° and 37° north. For most of the year the sun crosses the **southern** sky, so panels generally face **south**. Dense housing, overhead water tanks, staircase rooms (mumty), trees, dust and the monsoon all shape what a real Indian rooftop produces.

03 Fundamentals

SIMPLE

Irradiance is how strong the sunshine is right now. **Irradiation** is how much sunshine has fallen over a day — like rainfall collected in a bucket.

TECHNICAL

Irradiance: power per area, W/m² (bright noon sun ≈ 1,000 W/m²). Irradiation: energy per area, kWh/m² per day or year. **Peak sun hours (PSH)** = daily irradiation ÷ 1 kW/m².

PRACTICAL

5 kWh/m²/day = 5 peak sun hours — as if the sun shone at full 1,000 W/m² strength for 5 hours. MNRE: most of India receives **4–7 kWh/m²/day**.

DECISION

Peak sun hours × system kWp × a loss factor ≈ daily units. More sun, fewer shadows, better angle → more units from the same panels.

Term	Meaning
GHI	Global Horizontal Irradiance — sunlight on a flat, horizontal surface
POA / GTI	Plane-of-array (tilted) irradiance — sunlight on the panel's actual angle
Orientation (azimuth)	The compass direction the panels face
Tilt	The angle of the panels from horizontal
Shade-free area	Roof area with no shadow during productive hours, all year

04 Science: The Sun's Path Over India

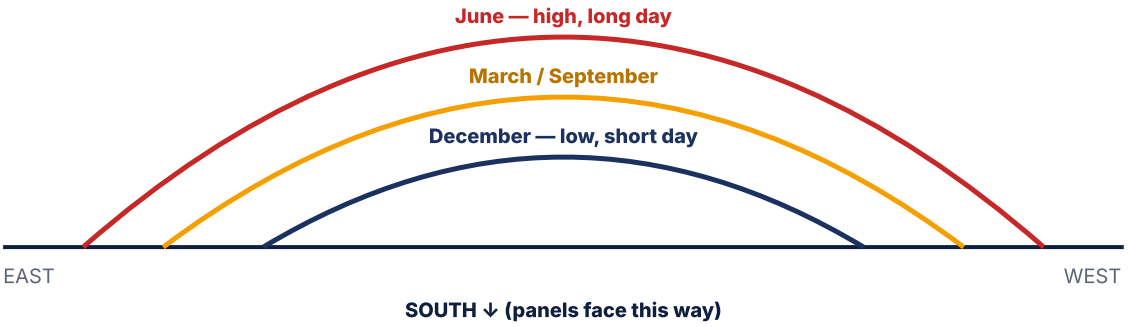


Figure 1. Simplified sun paths seen from a mid-Indian latitude, looking south. The December path is lowest — so December decides your shade.

The sun's height at noon follows a simple rule:

$$\text{Noon sun elevation } \alpha = 90° - \text{latitude} + \text{declination}$$

Declination is the sun's seasonal angle: about **+23.45°** on 21 June, **0°** at the equinoxes (≈21 March, ≈23 September) and **−23.45°** on 21 December.

City (latitude)	21 Dec	Equinox	21 June
Delhi (28.6° N)	38.0°	61.4°	84.9° (south)
Jaipur (26.9° N)	39.7°	63.1°	86.6° (south)
Pune (18.5° N)	48.1°	71.5°	≈85° (slightly **north**)
Chennai (13.1° N)	53.5°	76.9°	≈80° (**north**)

WHAT CHANGES IN INDIA? — THE SUMMER SUN GOES NORTH IN THE SOUTH

South of the Tropic of Cancer (≈23.4° N), the noon sun passes overhead in summer and then shines slightly from the **north** for some weeks around June. South-facing panels still collect the most energy over the **whole year**, but very steep south tilts are less helpful in southern India.

Tilt and orientation

ORIENTATION

- **South** — best yearly output in India.
- **East or west** — lower yearly output; west shifts generation to the afternoon.
- **East-west split** — flatter daily curve, useful on some roofs.

TILT

- A fixed tilt **close to your latitude** generally maximises yearly output.
- Flatter angles favour summer; steeper favour winter.

- **North** — avoid on fixed tilted arrays.

- Very flat panels collect more dust and drain rainwater poorly.
- Wind load and structure limit steep tilts — an engineer decides (R7).

Exact losses from non-ideal direction or tilt depend on location; ask your designer to quantify them with a simulation rather than accept a rule of thumb.

05 How Shade Works

SIMPLE

A shadow on a small part of a panel can reduce the output of much more than that part — sometimes the whole string.

TECHNICAL

Cells in series carry the same current; a shaded cell limits it. Bypass diodes route current around shaded sections, but that section's output is lost and the operating point shifts.

PRACTICAL

Water tanks, mumty rooms, parapets, trees, cables and neighbouring buildings are the usual culprits. Module-level electronics (optimisers, micro-inverters) can reduce shade losses at extra cost.

DECISION

First try to **avoid** shade by placement. Use shade-tolerant electronics only where shade truly cannot be avoided.

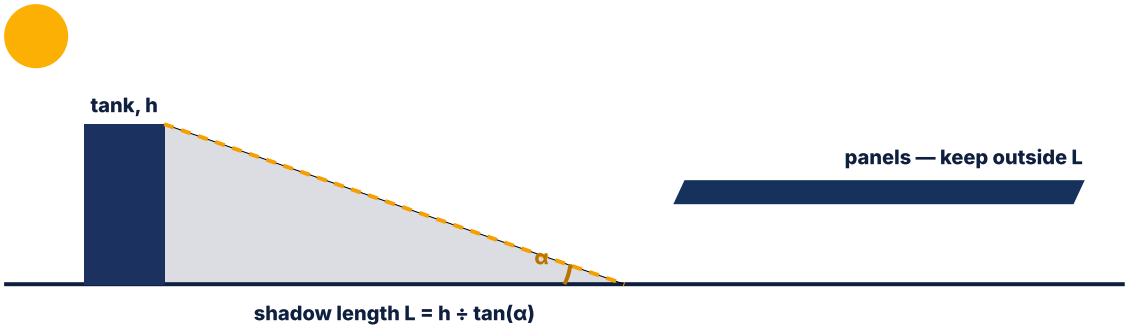


Figure 2. A taller object or a lower sun gives a longer shadow. Plan with the lowest sun of the productive day — December mornings and afternoons.

THE SON OF SURYA SHADE RULE

Aim for **no shade on any panel between 9 am and 3 pm on any day of the year** — check against the December sun. Shade outside those hours costs relatively little.

06 Indian Context

Son of Surya regional planning figures

Region / state	Planning figure (units per kWp per day)	≈ units per kWp per year
Rajasthan	4.1	1,490
Gujarat	3.8	1,400
Andhra Pradesh · Telangana	3.7	1,360
Maharashtra · Madhya Pradesh	3.6	1,320
Karnataka · Tamil Nadu · Delhi NCR	3.4	1,235
West Bengal · Assam & North-East	2.9	1,050
Other states, hills, heavy-monsoon coasts	Use the Son of Surya calculator or a site simulation	

Method: mid-points of published state yield benchmarks (1,225–1,750 kWh/kWp/yr, compiled from NREL, IRENA and NIWE data) reduced by 15% for real rooftop losses. Cross-checks: MNRE "4 to 5.5 units" per kWp on a clear sunny day; Tata Power-DDL Delhi example ≈1,200 units/kWp/yr. Conservative planning figures — not official values. Checked 7 Oct 2026.

Factor	Effect on your site
Monsoon	Cloud cover cuts output for weeks — typically June–September in most regions; Tamil Nadu and coastal Andhra also receive the north-east monsoon (Oct–Dec). Dry pre-monsoon months are usually the most productive.
Heat	Hot cells produce less; good airflow under panels helps.
Dust & haze	Dry months, farms, roads and construction raise soiling; winter haze in north India reduces sunlight.
Dense housing	Neighbouring buildings — including future construction — can shade your roof. Ask about plans next door.
Coastal sites	Salt mist corrodes metal; cyclonic winds load structures; coconut palms grow tall and cast moving shade. Specialist design needed (R8).
Roof space	MNRE: about **10–12 m² of shadow-free area per kWp** for grid-connected rooftop systems.

GLOBAL PERSPECTIVE

In Germany (≈48–55° N) the sun is much lower; panels often tilt more steeply and yields are lower per kWp. In the southern hemisphere (e.g. Australia), panels face **north**. Never copy foreign tilt or direction advice without adjusting for latitude.

07

Practical Application — The 20-Minute Site Walk

1. **Find south.** A phone compass is enough for a first look. Note which roof edges face south, east and west.
2. **Mark obstacles** on a rough sketch: tanks, mumty, parapets, dish antennas, trees, neighbouring buildings. Note each one's height.
3. **Check shadows at 9 am and 3 pm** — ideally in winter. In other seasons, use Calculation C3 to estimate December shadows.
4. **Measure the shade-free area** in m². Divide by 10–12 to estimate the maximum kWp the roof can hold.
5. **Look at the roof itself**: type (RCC slab, metal sheet, tiles), leaks, cracks, access for cleaning, and a safe route for cables.
6. **Ask about the future**: new floors, trees growing, neighbours building.

SAFETY

Do the walk only on a safe, flat roof with parapets. Do not climb sloping roofs, sheet roofs or water tanks. Structural checks of load and wind are for a qualified engineer.

08 Calculations

C1 Peak sun hours to daily units

FORMULA	$\text{Units/day} \approx \text{kWp} \times \text{PSH} \times \text{PR}$
VARIABLES	PSH = peak sun hours on the panel plane · PR = performance ratio (all losses), a fraction
ASSUMPTIONS	PSH 5.0 (within MNRE's 4–7 kWh/m²/day range) · PR 0.75 (a typical learning assumption)
WORKING	1 × 5.0 × 0.75 = **3.75 units per kWp per day**
INTERPRETATION	Close to the regional planning figures (2.9–4.1) — a useful cross-check on any vendor estimate.

C2 Noon sun elevation

FORMULA	$\alpha = 90° - \text{latitude} + \text{declination}$
WORKING	Delhi, 21 Dec: 90 − 28.6 − 23.45 = **37.95°**. Chennai, 21 Dec: 90 − 13.1 − 23.45 = **53.45°**.

C3 How long is the shadow?	
FORMULA	`L = h ÷ tan(α)`
ASSUMPTIONS	A 3 m high water tank on a flat roof
WORKING	Delhi, December noon: 3 ÷ tan 37.95° = 3 ÷ 0.780 = **3.85 m** (pointing north). Chennai, December noon: 3 ÷ 1.351 = **2.22 m**. At **9 am** in December the sun is lower — about 22° high in Delhi and 33° in Chennai, in the south-east — so the shadow is about **7.3 m** (Delhi) and **4.7 m** (Chennai), pointing north-west.
INTERPRETATION	Keep panels out of the north, north-west and north-east zones behind tall objects — further in north India than in the south.

C4 Roof capacity and yearly units	
FORMULA	`Max kWp ≈ shade-free area ÷ 11 m²  ·  Units/yr = kWp × regional figure × 365`
ASSUMPTIONS	40 m² shade-free area in Pune (Maharashtra 3.6) · 11 m² per kWp (middle of MNRE's 10–12)
WORKING	40 ÷ 11 = **3.6 kWp** → 3.6 × 3.6 × 365 = **4,730 units/year**
INTERPRETATION	If your yearly use is 2,800 units, the roof is not the limit — your consumption and DISCOM rules are.

09 Indian Case Study

ILLUSTRATIVE EXAMPLE — not a real customer or project

A terrace in Jaipur (26.9° N) has 55 m² of open roof, a 2.5 m water tank on the south edge and a mumty on the east. A vendor proposed panels across the whole roof in March. The owner checked:

- December noon elevation 39.7° → tank shadow 2.5 ÷ tan 39.7° ≈ **3.0 m** northwards at noon — and much longer at 9 am and 3 pm.
- The panel rows directly north of the tank would be shaded on winter mornings and afternoons — and in a series string, those panels can drag down the rest.
- Shade-free area after excluding the tank and mumty zones: about 38 m² → ≈ 3.4 kWp maximum → 3.4 × 4.1 × 365 ≈ **5,090 units/year** at Rajasthan's planning figure.

OUTCOME

The owner asked for a layout drawing marking the December shadow zone, a revised design avoiding it (or moving the tank), and a generation estimate with shading losses shown.

10 Cost & Economics

Sunlight is free, but **bad sunlight is expensive**: every panel you pay for in a shaded spot costs the same as one in full sun but produces less for decades. Paying slightly more for a raised structure, a relocated tank or shade-tolerant electronics can be worth it — but only when a simulation shows the gain. Ask for the yearly units *with and without* the fix.

11 Common Mistakes

Mistake	Reality
Checking shade at summer noon	Check 9 am–3 pm in December
Facing panels to match the building's edge	Aim for south where the structure allows
Copying a tilt from another country	Tilt depends on latitude
Ignoring small shadows (cables, poles)	Even small shade can cut string output
Forgetting future buildings and trees	Ask neighbours and plan for growth
Assuming all roof area is usable	Exclude shade, access paths and edges
Using a sunny-day figure for the year	Use regional yearly averages

12 Buyer Protection

- [ ] Ask for a **layout drawing** showing panel positions, obstacles and shadow zones
- [ ] Ask whether a **shading analysis** was done, and for which months and hours
- [ ] Ask for the **direction and tilt** used, and the expected yearly kWh with shading losses
- [ ] Ask how the structure handles **wind**, and for anti-corrosion treatment on coastal sites
- [ ] Ask about **cleaning access** and a safe walkway

13 Practical Checklist — Site Walk

- [ ] City / latitude: _____ · Region figure: _____
- [ ] Roof type: RCC / sheet / tile · Condition OK?
- [ ] South direction marked
- [ ] Obstacles listed with heights
- [ ] December 9 am–3 pm shadows estimated (C3)
- [ ] Shade-free area: _____ m² → max _____ kWp
- [ ] Expected units/year: _____
- [ ] Coastal / high-wind / heavy-dust site? _____
- [ ] Future shade risks noted
- [ ] Safe access for cleaning and maintenance

SON OF SURYA FREE TOOLS

Put your area, region and bills into the **Son of Surya Solar Calculator**, and record your site photos and measurements in the free **Solar Project Passport** at sonofsurya.com.

14 Problem Solving

Level	Problem	Solution
EASY	A site receives 5.5 kWh/m²/day. How many peak sun hours?	**5.5 hours** (5.5 ÷ 1 kW/m²)
INTERMEDIATE	Pune (18.5° N): December noon shadow of a 2 m parapet-mounted tank?	α = 90 − 18.5 − 23.45 = 48.05°; L = 2 ÷ tan 48.05° = 2 ÷ 1.113 = **1.80 m**
REAL-WORLD	Kolkata home: 27 m² shade-free area; yearly use 2,400 units. What size fits the roof, and what does it produce?	Roof limit: 27 ÷ 11 ≈ **2.5 kWp**. Output: 2.5 × 2.9 × 365 ≈ **2,650 units/year** — enough for the stated use on a yearly basis. Size to need: 2,400 ÷ 365 ÷ 2.9 ≈ 2.3 kWp.

15 FAQ

Q1 Which direction is best in India?
South, for the most yearly output.

Q2 What tilt should I use?
Near your latitude for maximum yearly output; your designer balances wind, dust and structure.

Q3 Do panels work on cloudy days?
Yes, but output falls a lot under thick cloud.

Q4 Is a little shade acceptable?
Only outside 9 am–3 pm, or with design measures. Small shadows can cost much more than their size.

Q5 How much roof do I need per kW?
About 10–12 m² of shadow-free area (MNRE).

Q6 Can I install on a sheet roof?
Often yes, with suitable clamps and a structural check.

Q7 Does heat reduce output?
Yes — hot cells produce less. Airflow helps.

Q8 Should I raise panels above my tank?
Sometimes. An elevated structure must be designed for wind and load by an engineer.

Q9 Which months produce most?
Usually the dry, clear pre-monsoon months; least in cloudy monsoon months.

Q10 Are east-west systems bad?
Not bad — they produce less in total but spread output across the day.

16 Glossary

Azimuth — compass direction panels face.

Bypass diode — routes current around shaded cells.

Declination — sun's seasonal angle (±23.45°).

GHI — sunlight on a horizontal surface.

Irradiance — sunlight power, W/m².

Irradiation — sunlight energy, kWh/m².

Mumty — staircase room on a roof.

Optimiser / micro-inverter — panel-level electronics that limit shade losses.

POA / GTI — sunlight on the tilted panel plane.

PSH — peak sun hours per day.

PR — performance ratio: actual ÷ ideal output.

Soiling — dirt and dust on panels.

Solar elevation — sun's height above the horizon.

Tilt — panel angle from horizontal.

17 What To Do Next

R1 ✓ → R2 ✓ → R3 Sun & Site ✓ → SOS-01 Starter → Free Calculator

18 Sources

Source	Used for	Checked
MNRE — Solar Overview (mnre.gov.in)	4–7 kWh/m²/day	7 Oct 2026
MNRE Rooftop Solar FAQ via PIB (28 Apr 2022)	10–12 m² per kWp; 4–5.5 units clear day	7 Oct 2026
Tata Power-DDL rooftop example (Dec 2020)	Delhi yield cross-check	7 Oct 2026
Qbits Energy state yield benchmarks (NREL/IRENA/NIWE compiled)	Regional planning figures	7 Oct 2026
Standard solar geometry (declination ±23.45°, elevation formula)	Sun angles and shadows	reference

19 Disclaimer

Published by Son of Surya for general education; not an engineering design or site survey. Sun-angle and shadow results are simplified (flat ground, solar time, idealised geometry). Regional planning figures are Son of Surya estimates, not official values. Examples marked ILLUSTRATIVE EXAMPLE are invented. Roof and structural assessment, wind design and electrical work require qualified professionals. Do not climb roofs or structures unsafely.

Son of Surya

India’s structured solar knowledge library

Free foundation guides, practical calculators and honest, step-by-step knowledge — so every Indian home, farm and business can understand solar before buying it.

UNDERSTAND → CALCULATE → VERIFY → COMPARE → DECIDE → ACT

FREE TOOL

Son of Surya Solar Calculator

Enter your bills and location. See system size, yearly units and savings — with every assumption shown.

FREE TOOL

Solar Project Passport

Keep your solar project’s key numbers, documents and warranties in one place. An engineer-verified version is available.

FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

SPECIALISED GUIDES — RELEASING IN BATCHES

- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
- **Professional** — wiring, installation, EPC

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